



How can firms leverage responsible artificial intelligence to build competitive advantages? The role of supply chain justice and complexity

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ABSTRACT

In highly interconnected supply chains, firms are increasingly adopting artificial intelligence (AI) to improve their forecasting, resource allocation, and responsiveness. However, opaque and bias-prone algorithms can obscure decision-making processes and undermine collaboration between supply chain partners. Responsible AI, in which AI is designed, developed, and deployed in such a way that ensures its ethical conduct, transparency, and alignment with human values, offers a potential remedy, but its relational effects in supply chains remain unclear. Drawing on organizational justice theory, we test whether supply chain justice, encompassing both distributive and procedural justice, mediates the responsible AI–competitive advantage link, with supply chain complexity as a boundary condition. Survey data from 218 Chinese firms show that responsible AI fosters both distributive and procedural justice, which in turn facilitates firms' competitive advantages. Further, we identify an asymmetric moderating effect: high supply chain complexity weakens the mediating effect of distributive but not procedural justice. Our study advances the research on AI-enabled supply chains by identifying how ethical AI practices can foster firm performance via supply chain justice and when supply chain complexity weakens this effect. Practically, our findings suggest that in complex supply networks, ensuring transparent and consistent decision-making processes provides a more robust mechanism for sustaining competitive advantages compared with focusing only on outcome allocation.

1. Introduction

In today's rapidly evolving global markets, supply chains have become increasingly complex networks of interconnected firms (Lee et al., 2025) in which artificial intelligence (AI) plays a critical role in driving operational efficiency and strategic decision-making. AI enables firms to forecast their demand more accurately, optimize their resource allocation, and improve their responsiveness, creating significant competitive advantages (Benbya et al., 2021; Roberts and Candi, 2024). However, the widespread adoption of AI also introduces new relational challenges (Benbya et al., 2020; Fosso Wamba et al., 2023). The “black box” nature of AI systems and their potential embedded biases can obscure how decisions are made and raise concerns about fairness among supply chain partners (Agrawal et al., 2020). When partners

perceive decisions as unfair or opaque, this can erode trust, undermine collaboration, and diminish the expected firm performance benefits brought by investments in AI (Sartori and Bocca, 2023; Schoenherr et al., 2023).

In response to this challenge, the concept of “responsible AI” has emerged as a strategic approach to the design, development, and deployment of AI systems to ensure they operate ethically, promote social good, and align with fundamental human values (Dignum, 2019; Laine et al., 2024). Within supply chains, responsible AI involves partner-facing practices such as bias mitigation in AI-driven resource allocation and partner selection, explainable AI outputs that are accessible to stakeholders, and robust data protection mechanisms (Kumar et al., 2023). While responsible AI holds great promise, its outcomes will depend largely on how supply chain partners perceive and respond to

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